

Name: _____ Date: _____

Period: _____

1. The table below shows inputs and outputs for a function.

x	$f(x)$
0	5
1	10
2	20
3	40

Exponential

Check: **Ratios:** $10/5 = 2$, $20/10 = 2$, $40/20 = 2$ — **constant ratio = 2.**

Write the formula: $f(x) = 5 \cdot 2^x$

2. Write the exponential function through (0, 3) and (2, 75).

Find the base b : $b = \left(\frac{75}{3}\right)^{1/2} = \sqrt{25} = 5$

Write the function: $f(x) = 3 \cdot 5^x$

3. **True or False:** A linear function and an exponential function can each be uniquely determined by knowing just two input-output pairs.

True

Explain: **Two points determine a unique slope (for linear) or a unique base and initial value (for exponential), so both function types are fully specified by two input-output pairs (EK 2.2.B.3).**